PRANAY DHANKE

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GitHub | LinkedIn | LeetCode | Portfolio | DockerHub

PROFESSIONAL SUMMARY

Computer Science Engineering student with hands-on experience designing scalable backend systems, distributed microservices, and real-time collaborative applications using Go, Node.js, TypeScript, and Next.js. Seeking SDE / Full-Stack / Backend internship opportunities.

TECHNICAL SKILLS

- **Languages:** Go, JavaScript, TypeScript, Java, Python
- **Backend:** Go (Gin), Node.js, Express.js
- **Frontend:** Next.js, React.js, Tailwind CSS, shadcn/ui, Redux
- **Databases:** PostgreSQL, MongoDB, Redis, Firebase
- **Infrastructure & Tools:** Git, GitHub, Docker, Kubernetes

PROFESSIONAL EXPERIENCE

Full-Stack Web Developer Intern — June 2024 – July 2024

Unified Mentor Pvt. Ltd.

- Developed a full-stack e-Governance platform for government services and welfare schemes for rural communities.
- Digitized manual application and verification workflows, reducing paperwork by approximately 90% and improving operational efficiency.
- Built responsive interfaces and integrated backend APIs to streamline scheme management and citizen service delivery.

PROJECTS

E-Editor – Real-Time Collaborative Code Editor — GitHub

Next.js, TypeScript, Express.js, Socket.IO, WebRTC, Yjs, Monaco Editor, Redis, MongoDB, Clerk, BullMQ, Docker, Kubernetes

- Engineered a real-time collaborative IDE using Yjs (CRDT) for conflict-free multi-user document synchronization and Monaco Editor, supporting 20 concurrent users per room with 1.15s P95 room-join latency and zero errors.
- Load-tested 100 concurrent users across 5 parallel rooms (20 users/room) via Redis Pub/Sub, achieving 1.07s P95 room-join latency with zero errors across the workload.
- Stress-tested single-room capacity to 1,000 concurrent connections, identifying code-update broadcast fan-out as the bottleneck as P95 latency reached 22.3s, establishing a 20-user-per-room production capacity target.
- Implemented Redis Pub/Sub with the Socket.IO adapter for cross-instance event propagation, supporting real-time presence, chat, and WebRTC-based audio/video collaboration.
- Offloaded room lifecycle cleanup and Yjs snapshot persistence to BullMQ workers, provisioned isolated Docker containers with CPU/memory limits for code execution, and authored Kubernetes Helm charts for repeatable deployments.

Smart-AgriWaste Platform — Website | GitHub

Next.js, Express.js, MongoDB, Redux, Next-Intl, Clerk, ImageKit, Socket.IO, OneSignal, Go (Gin)

- Built a full-stack agricultural marketplace enabling farmers to list categorized agricultural waste and connect with buyers for browsing, negotiation, and purchasing.
- Developed a Go + MongoDB waste-management API that maps user-provided agricultural-waste inputs to relevant management and utilization solutions through structured REST endpoints.
- Implemented secure Clerk authentication and real-time buyer-seller communication with Socket.IO, along with multilingual support (English, Hindi, Marathi) via Next-Intl.
- Added voice accessibility (speech input, text-to-speech), OneSignal push notifications, and PWA support to improve usability in low-connectivity rural environments.

Distributed Event-Driven Order Management System — GitHub

Go, Gin, RabbitMQ, PostgreSQL, Redis, Docker

- Built an event-driven Order Management System with Order, Inventory, Payment, and Notification microservices communicating asynchronously through RabbitMQ topic exchanges.
- Implemented Saga orchestration, Outbox pattern, idempotent consumers, retry mechanisms, and dead-letter queues to maintain consistency and prevent duplicate processing during service failures.
- Ensured concurrency-safe inventory updates using PostgreSQL transactions with row-level locking, preventing overselling under simultaneous purchase requests.
- Containerized all services with Docker and implemented correlation-ID-based distributed tracing for end-to-end request tracking across asynchronous service boundaries.

EDUCATION

Sipna College of Engineering and Technology — 2023 – 2026
B.E. in Computer Science and Engineering — CGPA: 8.14/10

Government Polytechnic, Arvi — 2020 – 2023
Diploma in Computer Science and Engineering — 79.20%